Stem & Leaf Diagrams

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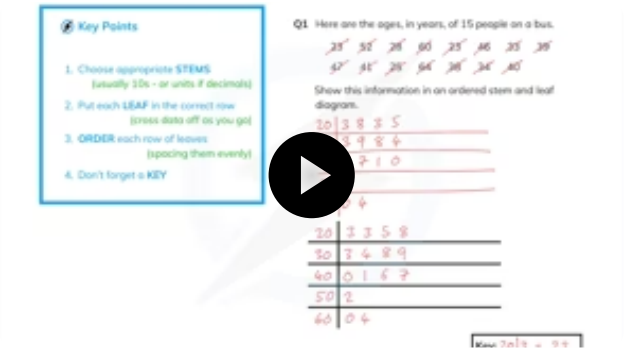

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Stem & leaf diagrams

What is a stem-and-leaf diagram?

Help me understand

- A **stem-and-leaf diagram** is a simple way to display an **ordered list** of data using **digits**
- Two-digit numbers are split into a tens digit (the **stem**) and a units digit (the **leaf**)
  - 25 becomes 2 | 5
  - The stem is written **vertically** and the leaves are written **horizontally** (in order)

- The following diagram shows the ages below
  - 11, 18, 20, 21, 25, 28, 29, 35, 36, 40

	Age
1	1 8
2	0 1 5 8 9
3	5 6
4	0

Key: 1|8 means 18 years old

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The **key** shows how values are **formed** from digits
  - It should include **units**
- Other keys are possible
  - 2 | 5 represents 2.5 degrees
  - 2 | 5 represents 2500 people

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How do I find the median from a stem-and-leaf diagram?

- The **median** is the **middle** number
  - Data values are already in **order**
  - You can cross out numbers from the beginning and end until they meet in the middle
    - Remember that the **highest** number will be at the **end** of the last stem
    - If two numbers remain in the middle, find the **midpoint** between them

Examiner Tips and Tricks

A common mistake when finding the median is to write down the leaf only (median = 6) rather than the stem and leaf (median = 26).

Worked Example

A hospital is investigating a new drug that claims to reduce blood pressure.
The reductions in blood pressure, measured in mmHg (millimetres of mercury), for 11 patients are shown below.

12 31 24 18 21 34 40 19 23 17 16

(a) Draw a stem and leaf diagram to show these results.

Answer:

Split each value into its tens digit (stem) and units digit (leaf)
The values are not yet in order

	Blood pressure reduction

3	1 4
4	0

Put the values in order
Write down the key

	Blood pressure reduction
1	2 6 7 8 9
2	1 3 4
3	1 4
4	0

Key: 1\|2 means a blood pressure reduction of 12 mmHg

(b) Use your stem and leaf diagram to find the median blood pressure reduction.

Answer:

Find the middle value
This is the 6th patient

	Blood pressure reduction
1	2 6 7 8 9
2	1 3 4
3	1 4
4	0

The median has a leaf of 1 and a stem of 2

The median is 21 mmHg

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